

# TAPAS MISHRA

Haldwani, Uttarakhand

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## Education

### Graphic Era Hill University

Bachelor of Technology (B.Tech) in Computer Science (CGPA: 8.1)

Bhimtal, Uttarakhand

2024 – Present

## Professional Summary

Motivated Computer Science undergraduate with strong foundations in **Data Structures** and **Web Development**. Experienced in building scalable web applications and AI-driven systems. Passionate about problem-solving and applying machine learning to solve real-world challenges.

## Relevant Coursework

- Data Structures
- Software Methodology
- Algorithm Analysis
- Database Management
- Internet Technology
- Computer Architecture

## Projects

### Proactive Cold Chain Alert System | Python, ML, Time-Series, Hash Chains

- Developed a proactive cold chain monitoring system that predicts vaccine temperature breaches up to **2 hours in advance** using a time-series machine learning model with **75%+ prediction accuracy**.
- Designed an optimized prediction algorithm capable of processing sensor data in **<2 seconds** on **1 GB RAM** devices while minimizing false positives caused by sensor noise.
- Implemented a tamper-evident audit logging system using **hash chains** to ensure secure, immutable, and traceable temperature records.
- Built an early warning mechanism that generates actionable alerts for healthcare workers **1–2 hours before** potential cold chain failures.

### Employee Management System (EMS) | React.js, Vite, JavaScript, Context API

- Developed a role-based Employee Management System with dedicated **Admin and Employee dashboards** for efficient workforce management.
- Implemented secure authentication, authorization, and protected routes using the **React Context API** for role-based access control.
- Built reusable and modular **React components**, improving maintainability, scalability, and code reusability.
- Optimized frontend performance using **Vite**, enabling faster development, reduced build times, and improved user experience.

### My Civic Issues | React.js, Firebase, SQL, JavaScript, CSS

- Developed a web-based platform that enables users to report, track, and monitor civic issues through an intuitive interface.
- Integrated **Firebase services** for authentication, data management, and image uploads, with **SQL support** for structured data handling.
- Implemented issue reporting features including **location mapping**, image uploads, and detailed complaint forms for accurate issue documentation.
- Designed a responsive and mobile-friendly user interface using **React.js** and **CSS** to provide a seamless experience across devices.

## Technical Skills

**Languages:** C++, JavaScript, Python

**Frontend:** React.js, HTML, CSS, Tailwind CSS

**Backend/Databases:** Node.js, Express.js, Django, MongoDB, Firebase, SQL, SQLite

**Cloud & Tools:** AWS (EC2, S3, IAM), Git, GitHub, Vercel, VS Code

**Core Concepts:** Data Structures & Algorithms, REST APIs, Authentication

## Extracurricular & Achievements

- **Hack the Winter Finalist:** Recognized as a finalist for **FrostLedger**, demonstrating excellence in **ML-driven** health solutions and time-critical proactive alerts.
- **LeetCode:** Achieved a competitive contest rating of **1610+ (Top 30% globally)** and successfully solved **350+ problems** with a strong core focus on complex algorithms.
- **GraFest 2025:** Secured **1st Place** in Survey Haul (Treasure Hunt) track at GEHU Bhimtal, demonstrating leadership, sharp teamwork, and rapid problem-solving skills.
- **We Code Club:** Dedicated and active member of GEHU Bhimtal's **We Code Club**, continuously participating in university-level hackathons and technical codeathons.